A CRITERION FOR A FRACTION TO BE THE SUM OF THREE UNIT FRACTIONS

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Extending the (in)famous Erdős-Straus conjecture, Schinzel [4] conjectured that any fraction $a/b \in (0,1)$ can be written as the sum of three unit fractions, as long as b is sufficiently large in terms of a. This is known in the case that $b+1$ has a prime factor $q \equiv -1 \pmod{a}$, because $b+1 = dq$ and $q+1 = am$ imply

$$\frac{a}{b} = \frac{1}{bm} + \frac{1}{dm} + \frac{1}{bdm}. \tag{1}$$

This identity already appears (in a more general form) in [3], and Vaughan [5] proved that, for a fixed a, almost all b lead to three-term unit fraction decompositions of a/b.[1] Another quick application of (1) that has, as far as the author is aware, not explicitly appeared in the literature before, gives the following result.

Theorem 1. *For all positive integers a and all primes p there exists a positive integer $r = r(a,p)$ such that every fraction $a/b \in (0,1)$ whose denominator b is divisible by p^r can be written as the sum of three unit fractions.*

Proof. With a and p given, we note that it is sufficient to show that there exists an r such that a/p^r can be written as the sum of three unit fractions. We may furthermore without loss of generality assume $p \nmid a$: if $a = a_0 p^s$, then a decomposition of a_0/p^r implies one for a/p^{r+s}.

Let us first assume $p = 2$, in which case we let q be a prime with

$$q \equiv -1 \pmod{a} \qquad \text{and} \qquad q \equiv 3 \pmod{8}.$$

Such a prime exists by the Chinese remainder theorem and Dirichlet's theorem, and it implies

$$\left(\frac{p}{q}\right) = \left(\frac{2}{q}\right) = -1.$$

On the other hand, if $p > 2$, let q be a prime with

$$q \equiv -1 \pmod{4a}$$

for which $-q$ is a quadratic non-residue modulo p. Once again, such a q exists by the Chinese remainder theorem and Dirichlet's theorem, while quadratic reciprocity implies

$$\left(\frac{p}{q}\right) = \left(\frac{-q}{p}\right) = -1.$$

In either case we take $r = \frac{q-1}{2}$, so that Euler's criterion implies $q \mid (p^r + 1)$. As $q \equiv -1 \pmod{a}$ by construction, identity (1) now gives a three-term decomposition of a/p^r, finishing the proof. □

[1] In fact, such an 'almost all' result can even be extended to two-term decompositions, see e.g. [1] and [2].

A consequence of Theorem 1 is that a/b is the sum of three unit fractions whenever b is smooth and sufficiently large.

Corollary 2. *For all positive integers a and all y there exists a $b_0 = b_0(a,y)$ such that the fraction a/b can be written as the sum of three unit fractions for all y-smooth $b > b_0$.*

Proof. One can take

$$b_0(a,y) := \max\left\{a, \prod_{p\le y} p^{r(a,p)-1}\right\}.$$

Indeed, if $b > b_0$ is y-smooth, then $p^{r(a,p)} \mid b$ for some $p \le y$ by the pigeonhole principle, after which we can apply Theorem 1. □

References

[1] G. Hofmeister and P. Stoll, *Note on Egyptian fractions*, J. Reine Angew. Math. **362** (1985), 141–145.

[2] J.-J. Huang and R. C. Vaughan, *On the exceptional set for binary Egyptian fractions*, Bull. Lond. Math. Soc. **45** (2013), no. 4, 861–874.

[3] M. Nakayama, *On the decomposition of a rational number into "Stammbrüche"*, Tôhoku Math. J. **46** (1940), 1–21.

[4] W. Sierpiński, *Sur les décompositions de nombres rationnels en fractions primaires*, Mathesis **65** (1956), 16–32.

[5] R. C. Vaughan, *On a problem of Erdős, Straus and Schinzel*, Mathematika **17** (1970), 193–198.

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